

Task 9: Random Forest Classification Report

Mobile Price Prediction Classification Project | 20 September 2026

1. Business Requirement

The mobile-phone company requires a machine-learning solution that classifies phones into price categories using technical specifications such as RAM, storage, battery capacity, camera quality, screen resolution, processor features, connectivity, and other hardware characteristics.

2. Task Objective

Task 9 develops and trains a Random Forest classifier using the prepared training dataset. The model learns relationships between encoded mobile-phone specifications and the three defined price categories. The held-out testing dataset is reserved for the separate Task 10 evaluation.

3. Methodology

The model was trained with scikit-learn's RandomForestClassifier. The training data contains 813 records, 26 numeric predictor features, and the Price_Category_Encoded target. The model uses 300 decision trees, random_state=42 for reproducibility, and n_jobs=-1 to use available CPU cores during training.

4. Training Summary

Measure	Result
Training records	813
Predictor features	26
Target column	Price_Category_Encoded
Target classes	3
Random Forest trees	300
Random state	42
Training diagnostic accuracy	99.75%

5. Training-Class Distribution

Price category	Class code	Training records	Percentage
Below 15,000	0	230	28.3%
15,000-29,999	1	294	36.2%
30,000 and above	2	289	35.5%

The training diagnostic accuracy describes how well the fitted model reproduces the data it has already seen. It is not a final measure of real-world performance; accuracy, precision, recall, F1-score, and the confusion matrix must be calculated on testing_dataset.csv in Task 10.

6. Feature Importance

Feature	Importance
numeric__Resolution_Megapixels	0.1124
numeric__Processor_Speed_GHz	0.1043
numeric__Resolution_Height	0.0931
numeric__RAM_GB	0.0737
numeric__Front_Camera_MP	0.0706
numeric__Total_Camera_MP	0.0694
numeric__Resolution_Width	0.0693
numeric__Charging_W	0.0509
numeric__Has_NFC	0.0479
numeric__Battery_per_Screen_Inch	0.0470

7. Deliverables

Implementation: random_forest_classification.py Trained model: random_forest_model.pkl Model metadata: random_forest_metadata.json Training input: training_dataset.csv

8. Conclusion

Task 9 is complete. A reproducible Random Forest classifier has been trained on the prepared mobile-phone specifications and saved for reuse. The model is ready for Task 10, where its predictions on the unseen testing dataset will be evaluated using classification metrics and a confusion matrix.

Generated from the project model and datasets by generate_task_9_report.py.